Memory–Logic Trade-off of Full-Cycle and Quarter-Wave Sine ROMs in a Fixed-Point AM Modulator on Artix-7

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***Abstract*—We present a fixed-point double-sideband full-carrier (DSB-FC) amplitude modulator written in Verilog, in which two numerically controlled oscillators (NCOs) generate the carrier and the message and two integer multipliers form the modulated output. The design is verified bit-exactly against an integer-only Python golden model over 13 configurations covering modulation indices from 0 to 1.5, DSB-SC operation and a wider message bandwidth; no sample of the 130 000 compared differs. We then compare two sine-table architectures, a full-cycle ROM and a quarter-wave ROM, for 8-, 10- and 12-bit table addresses on an xc7a35tcpg236-1 device at 100 MHz. The two architectures produce identical output at every address width, so the measured SFDR of 48.1, 60.7 and 71.9 dB depends only on the address width and follows $6.02\,AW$ dB. The quarter-wave table stores 75% fewer words; [measured LUT, BRAM and Fmax difference]. All figures are reproducible from the repository with a single command.**

***Index Terms*—amplitude modulation, direct digital synthesis, FPGA, quarter-wave ROM, fixed-point, Verilog**

I. Introduction

Amplitude modulation (AM) remains the simplest way to impress a low-rate message on a high-frequency carrier, and its envelope-detectable waveform makes it a convenient test case for digital transmitter design. In a digital implementation the carrier and the message are produced by direct digital synthesis (DDS) [1]: a phase accumulator addresses a sine table, and the output frequency is set by a single tuning word, so the same hardware serves any frequency without redesign.

The sine table dominates the memory cost of a DDS. Because a sine wave is symmetric, storing one quarter of a period is enough to rebuild the whole period, at the price of extra logic that mirrors the address and negates the sign [2], [3]. Whether this saving pays off on an FPGA, where small tables are mapped to distributed RAM and large ones to block RAM, is not obvious in advance. This paper answers the question for a complete AM modulator rather than for an isolated NCO.

The contributions are: (i) a parameterized Verilog AM modulator in which the ROM architecture and the address width are the only design variables; (ii) a bit-exact verification flow driven by an integer-only golden model and spectral checks; and (iii) post-implementation resource, timing and power figures for six ROM configurations, generated by a single script.

The scope is the modulator only. Demodulation is not implemented, and no hardware board is used: all results come from simulation and from Vivado synthesis and implementation targeting the xc7a35tcpg236-1 device.

II. Theory

A. Amplitude modulation

A DSB-FC AM signal is

$$s(t) = A_c\big[1 + \mu\, m(t)\big]\cos(2\pi f_c t), \qquad |m(t)| \le 1, \tag{1}$$

where f_c is the carrier frequency, A_c the carrier amplitude and $\mu = A_m/A_c$ the modulation index. For a single-tone message $m(t) = \cos(2\pi f_m t)$, (1) expands to

$$s(t) = A_c\cos(2\pi f_c t) + \tfrac{\mu A_c}{2}\cos\big(2\pi(f_c + f_m)t\big) + \tfrac{\mu A_c}{2}\cos\big(2\pi(f_c - f_m)t\big), \tag{2}$$

that is, exactly three spectral lines. The two sidebands are each $\mu/2$ of the carrier (-12.04 dB for $\mu = 0.5$), the occupied bandwidth is $B = 2f_m$, and only a fraction $\mu^2/(2+\mu^2)$ of the transmitted power carries the message [4].

The envelope $A_c[1+\mu m(t)]$ stays positive only for $\mu \le 1$. At $\mu = 1$ it touches zero (critical modulation), and for $\mu > 1$ it changes sign, so an envelope detector recovers a distorted message. This overmodulation is a property of the receiver; the transmitted product in (1) is still linear, so its spectrum keeps three lines. Setting $A_c = 0$ removes the carrier line and gives DSB-SC, for which the sideband-to-carrier ratio is undefined.

B. Fixed-point implementation

Samples are signed integers in which 2047 represents 1.0. The product of two such numbers carries the scale factor 2^{11} twice, so each multiplication is followed by an arithmetic right shift of 11 bits:

$$m_t = (A_m \cdot m) \gg 11, \quad e = A_c + m_t, \quad y = (e \cdot c) \gg 11, \tag{3}$$

with m and c the 12-bit message and carrier samples, e a 14-bit envelope and y a 16-bit output. An arithmetic right shift rounds toward $-\infty$, which the golden model reproduces with the Python » operator; any floating-point reference rounded at the end would differ by a few LSB and never match bit for bit.

TABLE I
DATAPATH WIDTHS.

Operation	Inputs (bits)	Raw product	After shift
$A_m \cdot m$	12×12	25	14
$e \cdot c$	14×12	26	16

C. Numerically controlled oscillator

A W-bit phase accumulator adds a frequency tuning word (FTW) every clock cycle, and the AW most significant bits address a sine table. With clock frequency f_{clk},

$$f_{out} = \frac{\mathrm{FTW} \cdot f_{clk}}{2^W}, \qquad \Delta f = \frac{f_{clk}}{2^W}. \tag{4}$$

For $W = 32$ and $f_{clk} = 100$ MHz the resolution is 0.023 Hz; the carrier (1 MHz) uses FTW $= 42\,949\,673$ and the message (20 kHz) FTW $= 858\,993$. Discarding $W - AW$ phase bits (phase truncation) adds spurious lines whose level improves by roughly 6 dB per address bit [2], which motivates sweeping $AW \in \{8, 10, 12\}$.

D. Full-cycle and quarter-wave tables

A full-cycle table stores 2^{AW} words of $\mathrm{round}\big(2047 \sin(2\pi(n + 0.5)/2^{AW})\big)$. The half-sample offset makes the table exactly antisymmetric, so the second quarter is the mirror image of the first and the second half is the first half negated. A quarter-wave table therefore stores 2^{AW-2} words; bit $AW - 2$ of the phase selects mirroring (address inversion) and bit $AW - 1$ selects negation. This removes 75% of the words but adds an address inverter and a sign negation to the read path.

III. Architecture

Fig. 1 shows the modulator. Two NCOs produce the carrier c and the message m; the envelope stage computes $e = A_c + (A_m m) \gg 11$ and the multiplier forms $y = (e\,c) \gg 11$. Each NCO is a 32-bit phase accumulator followed by a sine ROM addressed by the AW most significant phase bits. The ROM is selected by a single parameter (`USE_QUARTER_ROM`); every other module is shared, so any difference in resources between the two variants is caused by the ROM alone.

Both ROMs have exactly two register stages, so the NCO latency is identical. The quarter-wave ROM inverts the address bits when phase bit $AW-2$ is set, registers the magnitude and, in parallel, the sign bit $AW - 1$, and negates the registered magnitude in the second stage. The carrier is delayed by one register before the multiplier to match the registered envelope, so the output appears three clock cycles after the phase accumulators.

The unsigned amplitude A_m is extended with a leading zero and declared signed before the multiplication, so the product is correct in the negative half of the message.

IV. Verification

The reference is an integer-only Python model that reproduces (3) sample by sample. The Verilog testbench reads the stimulus parameters from the command line and writes the output samples to a text file. After reset the output starts with three warm-up samples; these are skipped explicitly and the remaining 10 000 samples are compared with the model. The acceptance criterion is bit-exact equality of all 10 000 samples.

Thirteen configurations are run (Table II). Cases 1–5 vary μ, case 6 sets $A_c = 0$ (DSB-SC), case 7 raises the message to 100 kHz, and cases 8–13 cross the three address widths with the two ROM architectures at $\mu = 0.5$. In addition, a Hann-windowed FFT of the 10 000 RTL samples (bin width 10 kHz, so every line falls on an exact bin) is used to check that the output contains the three lines of (2) at the expected frequencies with amplitude ratio $\mu/2$, and to measure the SFDR, defined as the strongest intended line over the largest FFT bin outside the guard bands of the intended lines, with DC counted as a spur.

V. Results

A. Bit-exact verification and spectra

All 13 configurations match the golden model with zero mismatching samples (Table II). For the default stimulus ($\mu = 0.5$) the envelope runs from 1000 to 2999 and the output from -2998 to 2997; the output peak lies one count below $A_c(1+\mu)$ because the right shift truncates toward $-\infty$. At $\mu = 1$ the envelope reaches 0, and at $\mu = 1.5$ it goes down to -999: the output keeps the three-line spectrum but the envelope detector would see the folded envelope of Fig. 2(b).

The measured spectrum has the three lines at 980, 1000 and 1020 kHz with no frequency error at the 10 kHz bin width, and each sideband lies within 2×10^{-4} of $\mu/2$ times the carrier in every case with a carrier (Fig. 2(e)). With $A_c = 0$ the carrier bin falls 60.1 dB below the sidebands (Fig. 2(f)), and raising the message to 100 kHz widens the occupied band to 200 kHz $= 2f_m$.

B. Full-cycle versus quarter-wave ROM

At every address width the full-cycle and quarter-wave designs produce the same 10 003 output words, so Table II and Fig. 3 show a single SFDR per AW: 48.07, 60.72 and 71.88 dB for $AW = 8$, 10 and 12, against $6.02\,AW = 48.2$, 60.2 and 72.2 dB. The largest spur lies at 45, 25 and 3 MHz respectively, far from the three AM lines. The quarter-wave table needs 768, 3 072 and 12 288 bits instead of 3 072, 12 288 and 49 152.

C. Resources, timing and power

Table III lists the post-implementation results. Each row is traceable to `results/rpt/AW<aw>_<arch>/`.

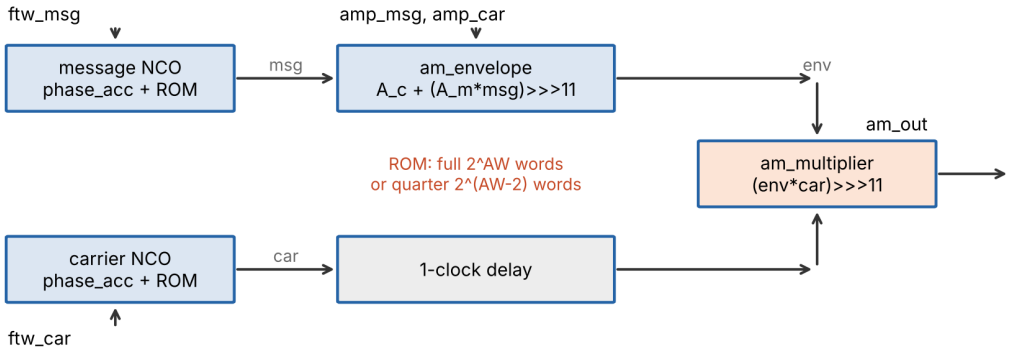


Fig. 1. Block diagram of the AM modulator. The carrier is delayed by one clock so that it reaches the multiplier in the same cycle as the registered envelope.

TABLE II
VERIFICATION RESULTS FOR THE 13 CONFIGURATIONS (10 000 SAMPLES EACH; SFDR FROM A HANN-WINDOWED FFT OF THE RTL OUTPUT).

#	AW	ROM	A_c	A_m	μ	Envelope range	Output range	Mismatches	SFDR (dB)
1	8	Full	2000	0	0	2000…2000	-2000…1999	0	48.2
2	8	Full	2000	500	0.25	1500…2499	-2498…2497	0	48.1
3	8	Full	2000	1000	0.5	1000…2999	-2998…2997	0	48.1
4	8	Full	2000	2000	1	0…3999	-3997…3996	0	48.1
5	8	Full	2000	3000	1.5	-999…4998	-4995…4994	0	48.1
6	8	Full	0	1000	–	-1000…999	-1000…999	0	47.9
7	8	Full	2000	1000	0.5	1000…2999	-2985…2984	0	48.1
8	8	Full	2000	1000	0.5	1000…2999	-2998…2997	0	48.1
9	8	Quarter	2000	1000	0.5	1000…2999	-2998…2997	0	48.1
10	10	Full	2000	1000	0.5	1000…2999	-2998…2997	0	60.7
11	10	Quarter	2000	1000	0.5	1000…2999	-2998…2997	0	60.7
12	12	Full	2000	1000	0.5	1000…2999	-2998…2997	0	71.9
13	12	Quarter	2000	1000	0.5	1000…2999	-2998…2997	0	71.9

TABLE III
POST-IMPLEMENTATION RESULTS, XC7A35TCPG236-1, 100 MHZ TARGET (OUT-OF-CONTEXT). $F_{max} = 1/(10\,\text{ns} - \text{WNS})$.

AW	ROM	Table bits	LUT	FF	LUTRAM	BRAM	DSP	WNS (ns)	F_{max} (MHz)	P_{dyn} (W)
8	Full	3 072	–	–	–	–	–	–	–	–
8	Quarter	768	–	–	–	–	–	–	–	–
10	Full	12 288	–	–	–	–	–	–	–	–
10	Quarter	3 072	–	–	–	–	–	–	–	–
12	Full	49 152	–	–	–	–	–	–	–	–
12	Quarter	12 288	–	–	–	–	–	–	–	–

VI. DISCUSSION

Equivalence of the two architectures. The quarter-wave ROM is a pure storage optimisation: it reproduces the full-cycle output sample for sample, so nothing is paid in signal quality, and SFDR is set by the address width alone. The roughly 6 dB per address bit gain is the price-performance knob; the memory saving is independent of it, a constant 75%.

Memory versus logic. [Compare LUT, LUTRAM/BRAM and F_{max} of the two architectures at $AW = 8$, 10 and 12 using Table III: where the fold logic outweighs the memory saved, and whether the negation shortens the maximum frequency.]

Limitations. The design has not been run on a board; results are simulation and Vivado out-of-context implementation. The message is a single tone. The constraint file defines only the 10 ns clock and sets false paths on all ports, so $F_{max} = 1/(10\,\text{ns} - \text{WNS})$ reflects register-to-register paths inside the modulator and not I/O timing. Power is the vectorless Vivado estimate. Each configuration was implemented once with one tool version.

DSB-SC needs no extra hardware: it is the same datapath with $A_c = 0$, and it saves the carrier power at the cost of a coherent demodulator.

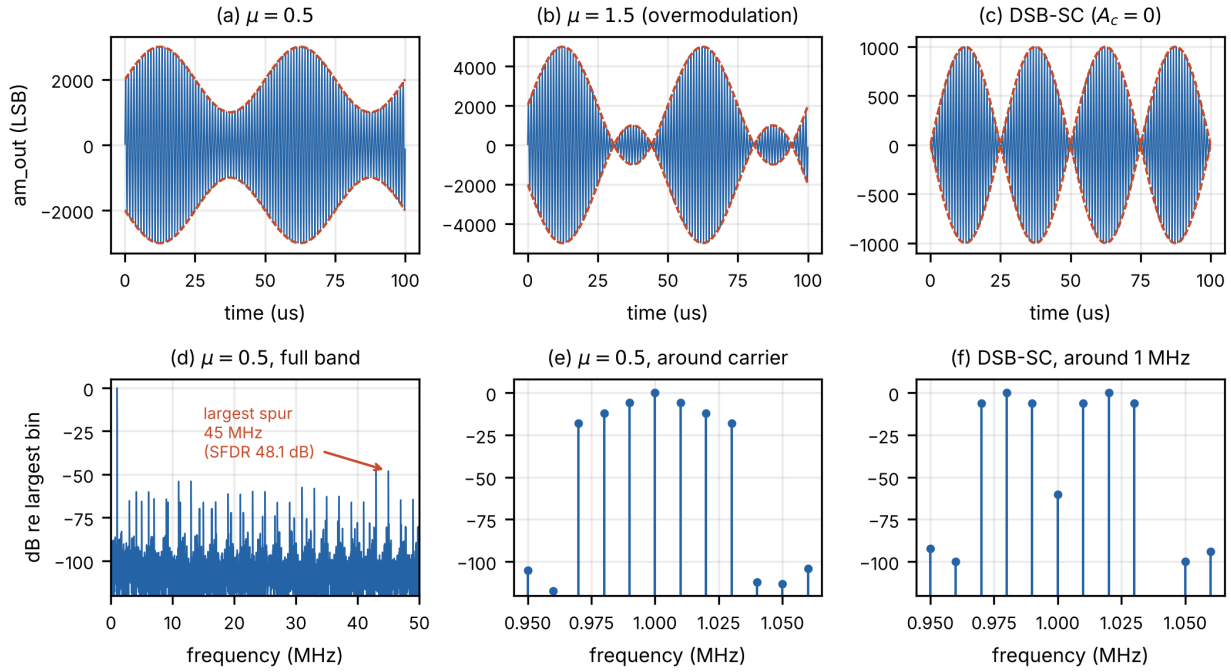


Fig. 2. RTL output (solid) with the signed envelope (dashed) for (a) $\mu = 0.5$, (b) $\mu = 1.5$ and (c) DSB-SC; spectra of the $\mu = 0.5$ output over the full band (d) and around the carrier (e), and of the DSB-SC output (f). Hann window, 10 000 samples.

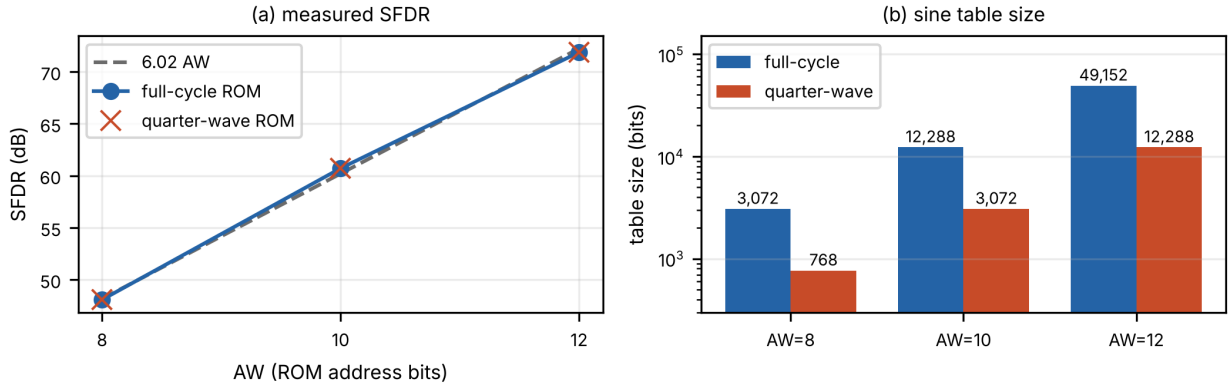


Fig. 3. (a) SFDR versus address width AW; the quarter-wave points coincide with the full-cycle points. (b) Sine table size.

VII. Conclusion and Future Work

A parameterized fixed-point AM modulator was verified bit-exactly against an integer golden model on 13 configurations, and its two sine-ROM variants were shown to be output-equivalent at three table sizes while the quarter-wave table stores 75% fewer bits. **[State the measured LUT, BRAM and F_{max} trade-off.]** Future work includes an envelope detector to close the modulate–demodulate chain, single-sideband modulation, pipelining to raise F_{max}, switching-activity-based power estimation and an AXI-Stream wrapper for a Zynq design.

References

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